Viral load kinetics: a way to assess cumulative effects of CMV on hematopoietic stem cell transplant recipients

M. Veronica Dioverti Prono MD* and Robin K. Avery MD,
Division of Infectious Diseases, Johns Hopkins University

*Corresponding author:
M. Veronica Dioverti Prono, M.D.
Assistant Professor of Medicine
Division of Infectious Diseases
Transplant/Oncology Infectious Diseases Program
Johns Hopkins University
600 N Wolfe Street, Carnegie Building 340
Baltimore, MD 21287
Phone: 443 287 6217
Fax: 410 955 0788
Email: mdiover1@jhmi.edu

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021

© The Author(s) 2021. Published by Oxford University Press for the Infectious Diseases Society of America. All rights reserved. For permissions, e-mail: journals.permissions@oup.com.

Cytomegalovirus (CMV) has been associated with significant morbidity in hematopoietic stem cell transplant (HSCT) recipients, both via direct effects (DNAemia, end organ disease) and indirect effects (increased risks for other viral and/or fungal infections, graft-versus-host disease)[1,2]. Predicting its impact on overall survival and non-relapse mortality post-HSCT has been the focus of many studies. However, understanding cumulative morbidity associated with infections is difficult, as time-to-first event analyses do not account for recurrent events, and estimates can be biased without controlling for death[3]. Moreover, CMV DNAemia remains a risk factor for overall and non-relapse-related mortality despite institution of a pre-emptive therapy approach[1].

In this issue of *the Journal of Infectious Diseases*, Stern et al. report a study of CMV outcomes in a large single-center HSCT population managed with pre-emptive therapy, and overcame these limitations by using the "average area under the curve" (AAUC) as a quantitative indicator of CMV load over time, a much more informative tool for cumulative morbidity than time-to-first DNaemia, or simply the percentage of patients who develop CMV DNAemia and/or end organ disease4. This tool incorporates not only the amplitude of DNAemia, but the total duration as well [4-7]. AAUC has been investigated for other dsDNA viruses and has been shown to correlate with increased overall-mortality post-HSCT; higher viral load (VL) at onset of DNAemia was associated with more persistent episodes, end-organ disease and higher mortality[5].

The authors demonstrate effectively that the outcomes associated with CMV are quite different, depending on the patient phenotype as well as the magnitude and duration of CMV DNAemia, and they classified patients based on episodes of CMV DNAemia before day 100 post-HSCT. The "non-controllers" (higher VL and longer CMV DNAemia) had significantly

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021

lower overall survival and higher non relapse mortality as compared with all others falling into lower quartiles ("controllers") as well as those defined as "elite controllers" and D-R-; these findings persisted even after adjusting for other covariates. Not surprisingly, HLA-mismatched and T cell-depleted (TCD) transplants were found to be predictive of "non-controller" status, while CMV-seropositive donor status was found to be protective. A beneficial effect of donor-seropositive status has been noted in other studies, particularly in unrelated-donor HSCT[8].

The median CD4+ count for both TCD and unmodified HSCT was lower for the non-controllers, suggesting perhaps that this group of patients has an overall poorer immune status compared to others, rendering them at risk for both CMV and non-CMV infections. Further supporting this is the fact non-controllers had more deaths due to infections in general when compared to other groups. Even though CMV replication significantly stimulates T cell function[9], it is also known to be immunosuppressive and increases risks for other infections[2]. Hence, it remains difficult to discern whether loweroverall survifal and higher non-relaps mortality in non-controllers is truly related to CMV, or mainly a reflection of a poor host in whom other factors may play a role.

Incorporating CMV viral load dynamics in a study population with a large sample size and including assessment of immune reconstitution (including CD4+counts), adds to a strong study design. Although the overall cohort is large, it is also quite heterogeneous. Almost half of patients had T cell depleted HSCT. Approximately 90% of all HSCT in this cohort were peripheral blood stem cell transplants (PBSCT); graft source may affect GVHD risk[10]. GVHD, in turn, is associated with increased risks for infections including CMV[11]. Indeed, in the study by Stern et al, acute GVHD episodes were more frequent in non-controllers;

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021

however, multivariable analyses showed that non-controllers remained at high risk for mortality even after adjusting for TCD and GVHD. This study did not consider CMV episodes beyond day 100; although rare, these may still influence non-relapse mortality at 1 year. The increasing use of letermovir prophylaxis may alter this dynamic, whereby CMV episodes might be more frequently late-onset in the future[12].

This study further supports the idea that CMV viral kinetics have a direct relationship with post-HSCT outcomes and should be incorporated as a tool when assessing treatment strategies. Despite the implementation and success of pre-emptive approaches to CMV DNAemia, it remains an independent risk factor for poor outcomes. This study identifies a group of CMV non-controllers as a particularly vulnerable group, deserving of further study in terms of preventive interventions. The potential benefits of universal prophylaxis for CMV post-HSCT with newer non-myelosuppressive agents, such as letermovir, should be assessed in larger clinical trials for this sub-group of non-controllers. Moreover, a predictive scoring tool could be developed to allow for more objective assessment of patients at higher risk, who may benefit from enhanced prevention strategies. In summary, Stern and colleagues have made an important contribution towards our understanding of cumulative morbidity from CMV, especially in the most vulnerable patients.

Footnotes: The authors have no conflicts of interest/funding related to the submitted work.

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021

References:

1. Green ML, Leisenring W, Xie H, et al. Cytomegalovirus viral load and mortality after haematopoietic stem cell transplantation in the era of pre-emptive therapy: a retrospective cohort study. Lancet Haematol 2016 Mar; 3(3): e119-127.
2. Nichols WG, Corey L, Gooley T, Davis C, Boeckh M. High risk of death due to bacterial and fungal infection among cytomegalovirus (CMV)-seronegative recipients of stem cell transplants from seropositive donors: evidence for indirect effects of primary CMV infection. J Infect Dis 2002; 185:273-282.
3. Dioverti MV, Huang CY, Morales M, Lu N, Cochran W, Gladstone D, Ostrander D, Avery R, Jones R, Ambinder R Marr K. Cytomegalovirus Reactivation in Haploidentical Blood or Marrow Transplant (BMT) Using Post-Transplant Cyclophosphamide (PTCy) Is Associated with Higher Non-Relapse-Related Mortality, Despite Rarity of Disease. Blood (2018) 132 (Supplement 1): 2096.
4. Stern A, Su Y, Dumke H, et al. CMV viral load kinetics predict CMV end-organ disease and mortality after hematopoietic cell transplant (HCT). J Infect Dis 2021;
5. Hill JA, Mayer BT, Xie H, et al. Kinetics of double-stranded DNA viremia after allogeneic hematopoietic cell transplantation. Clin Infect Dis 2018; 66: 368-375.
6. McBride JM, Sheinson D, Jiang J, et al. Correlation of cytomegalovirus (CMV) disease severity and mortality with CMV viral burden in CMV-seropositive donor and CMV-seronegative solid organ transplant recipients. Open Form Infect Dis 2019; 6(2): ofz003.
7. Gimenez E, Solano C, Vinuesa V, et al. Cytomegalovirus DNAemia burden and mortality following allogeneic hematopoietic stem cell transplantation: an area under the curve-based investigational approach. Clin Infect Dis 2018; 67(5):805-807.

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021

8. Ljungman P, Brand R, Einsele H, et al. Donor CMV serologic status and outcome of CMV-seropositive recipients after unrelated donor stem cell transplantation: an EBMT megafile analysis. Blood 2003; 102 (13): 4255-60.
9. Zhou W, Longmate J, Lacey SF, et al. Impact of donor CMV status on viral infection and reconstitution of multifunction CMV-specific T cells in CMV-positive transplant recipients. Blood 2009; 113 (25): 6465 – 6476.
10. Jagasia M, Arora M, Flowers MED, et al. Risk factors for acute GVHD and survival after hematopoietic cell transplantation. Blood 2012; 119(1): 296-307.
11. Miller HK, Braun TM, Stillwell T, et al. Infectious risk after allogeneic hematopoietic cell transplantation complicated by acute graft-versus-host disease. Biol Blood Marrow Transpl 2017; 23(3): 522-528.
12. Sperotto A, Candoni A, Gottardi M, et al. Cytomegalovirus prophylaxis vs. pre-emptive strategy: different CD4+ and CD8+ T cell reconstitution after allogeneic hematopoietic stem cell transplantation. Transplantation and Cellular Therapy 2021; doi: 10.1016/j.jtct.2021.03.003

Accepted Manuscript

Downloaded from https://academic.oup.com/jid/advance-article/doi/10.1093/infdis/jiab213/6232022 by Memorial Sloan Kettering Library user on 22 April 2021